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Ways of knowing: an introduction

LET'S START WITH EGGS. Most animal species have them, and so do many plants. If you so wished, you could obtain details on thousands and thousands of different animal eggs because naturalists have collected them, described them in print and filed away the specimens and the data. In any of the world's great natural history museums you could find out about the eggs of the praying mantis, the chameleon or the elephant. And for species with eggs of economic or medical importance – for parasitic flies, edible fish or for hens – the data available is overwhelming. How and why were all these specimens and data-sets collected and kept? Who filled the filing cabinets of the natural world?

And what of the eggs from which we as individuals began? What do we know about them? In 1600, experts would have said you came from a mixture of fluids – from the seminal fluid from your father which acted on the seminal fluid of your mother. Mammalian eggs were first described in the seventeenth century, and so were spermatozoa – the millions of tiny ‘animalcules’ in every drop of semen. What on earth, or in heaven, were they all for? We are so used now to moving-pictures of sperms, eggs and embryos, and of the late human foetus so ‘lifelike’ *in utero*, that we may forget how all those microscopical or hidden processes were made visible, how much ingenuity and interpretation those ‘natural’ images can hide. And that is just for the ‘outsides’ of eggs.

Suppose you want to ‘look inside’? Chemists can offer you reams about the chemical composition of the eggs you eat; they

know them down to the level of the chemical elements. Biologists can tell you that the fertilised egg is a cell which divides to give rise to the millions of cells in the adult, and that the nucleus of each cell contains chromosomes which become visible during cell-division. Embryologists know about the 'germ layers' of vertebrate embryos and the ways in which a fertilised egg becomes a microscopical, hollow ball of cells; how one side of the ball folds in, like a flattened tennis ball; how the two-layered structure then grows so that the inner wall becomes your gut lining, the outer wall your nervous system and your skin, and a third, intermediate (germ-) layer produces the rest of your body. Most of this wonderful 'protoplasmic origami' was analysed in the nineteenth century under the leadership of German professors fascinated by 'development' in embryos, lives and societies.

By the end of the nineteenth century, they were trying to intervene. What happened if you removed half of a divided egg? Would you get two whole-embryos, or two half-embryos, or something in between? What would that experiment tell you about the 'potencies' of the parts of the egg? In laboratories by the sea, where the professors could spend their summers playing with marine eggs, they began to learn how to control the development of eggs. And now we have artificial fertilisation, cloning and 'Dolly' the sheep. In the twentieth century, embryos were further analysed to reveal that chromosomes are made up of nucleic acids, constituting genes that are switched on and off during development. Structural and chemical analyses now converge at the level of macro-molecules (Hopwood, forthcoming; Jacob, 1974; Needham, 1959).

This book is about all the various *ways of knowing* which went into 'eggs as we know them'. It is about *natural history* – the describing and classification of things; about *analysing* into various kinds of elements, like germ layers, cells and chemical elements; and about *experimenting* to control phenomena and to systematically create novelties. I will try to show that much of science, technology and medicine (STM) can usefully be understood in terms of these three ways of knowing and their interactions, and that this approach also illuminates the *histories* of

STM. My account includes planets and stars, clouds and cuckoos, minerals and chemicals, machines and steam engines, patients and bacteria, vacuum tubes and radioactivity, electronics and pharmaceuticals, atom bombs and genetic engineering – even languages and societies. It may not be comprehensive – indeed, it is conspicuously weak on mathematics – but it is wide ranging. In the next sections I introduce my method and say more about this particular approach to a huge topic – the history of science, technology and medicine in the West from the sixteenth-century Renaissance to the present. But first I want to add two further dimensions, which in some ways ‘frame’ the three ways of knowing – natural history, analysis and experimentalism – that I have just introduced.

I want to show how ways of knowing were linked with ways of *production* – to ways of *making* things, or to ways of tending and mending (in agriculture and medicine), or of defending or destroying (in military sciences and technologies). I am interested in the ways in which knowledge is built into commodities or other products, such as new pharmaceuticals or new weapons. I use the term *technoscience* for technological projects which are heavily dependent on science (or vice versa), and I suggest that it was chiefly from the later nineteenth century that (some) academics, (some) industrialists and (some) state agencies began to create the networks for the systematic production of novelty which were increasingly characteristic of the twentieth century. Eggs, for example, are now big business, not just through the standardisation and mechanisation of hen-farming, but in high-tech firms which clone the eggs of high-value cattle, using all the techniques of experimental biology and working closely with universities and with government agencies.

Technoscience, I will argue, is central to our world – to our artefacts and to our ‘nature’. But we must read it aright. Why, for example, are the products of high-tech companies sold to us as ‘brands’, as symbols of (our) status, connoisseurship or modernity (Klein, 2000)? Their production may be technical beyond the understanding of most of humankind, but the appeal, manifestly, is a matter of ‘human meanings’. We can only fully

grasp these technologies if we understand the values; our appreciation of artefacts and of their invention and production must involve the study of *meanings*. But this is true for ‘nature’ also. *All* the worlds we know, whether through everyday life or ‘science’, have *meaning* for us. So they did also for the men and women who made our worlds and our understandings – but *their* understandings were not necessarily ours. To explore their *world-readings* and their purposes is part of the adventure of this book.

We can see that dimension easily enough by returning to eggs. When we watch a film of the developing embryo, it may suggest to us the possibility of modification and control – for better or worse. Few of us see there the glories of a divine creation unfolding before our eyes, though that attitude was the context and spur to much ‘anatomising’ from the Renaissance to the nineteenth century, and many of us still think it wise to cling to ‘*nature’s* way’, if only as ‘precaution’. Abnormalities of human development are now the subject of ‘dysmorphology’, a branch of medicine recording and analysing ‘monstrous births’ in the hope of finding genetic or environmental causes. They are not seen as punishments or portents in the ways which once motivated such investigations – but stricken parents may still feel the power of that understanding when they ask ‘why us?’ And when popular genetics suggests to us that much of an individual’s future is ‘pre-formed’, we might spare a thought for the seventeenth-century microscopists who saw tiny pre-formed beings hunched in the heads of sperms and thought they had solved one of the deepest quandaries of their (advanced) science. If God had made the world as a machine, how could one understand the ‘generation’ of new beings? Perhaps all beings had been ‘laid down’ at the Creation, stored one within another like Russian dolls. *That* was the meaning for them of the micro-mechanical men hidden in sperms (or maybe in eggs) and unfolding in the embryo. In a world of limited duration, perhaps all past and future generations had existed since creation.

And if you smile, spare a thought for the smiles of the future.

An outline of the method

My approach to the history of science, technology and medicine has four key features:

- 1 a long timescale: working through the history of the past 300 years;
- 2 breadth of scope: taking science, technology and medicine together – linking their history to other human histories;
- 3 dissecting science-technology-medicine into constituent elements – *ways of knowing*, with their different histories; and
- 4 presenting these *ways of knowing* as forms of *work* related to various ways of making and mending. In general, I use my ‘ways of knowing’ to bridge between esoteric, technical worlds and the worlds of ‘everyday’, both past and present.

Long timescale

Historians can be prissy about keeping to their periods. They worry that in studying events across centuries they may lose sensitivity to the peculiarities of particular times and places, and thus misread them. So we tend to specialise in particular periods and/or in the history of a discipline, e.g. the history of nineteenth-century physics. But when we work on a particular time and place, we can easily lose sight of the ‘big picture’, and the fact that we only know what something is when we know what it is *not*. To understand eighteenth-century medicine, say, and to communicate that understanding, we need to know how it differs from the medicine with which we live now. By knowing our contemporary medicine we can know more about the eighteenth century and vice versa. ‘Periods’ illuminate each other.

To be sure, we must be careful not to see eighteenth-century medicine in modern terms (the scholarly sin of ‘presentism’), or to focus exclusively on the roots of later developments to the exclusion of aspects whose importance has since faded (the sin called ‘Whiggism’), but in my view it is naive to pretend that we should or can forget our present categories. Could we do so, we

would be no more use than anthropologists ‘gone native’ and unable to relate to fellow investigators.¹ To do justice to the past and to *use it in the present*, we need broad frames in which to think comparisons. For such reasons, this book takes a long period, and also a very broad definition of ‘science’.

Wide scope

Most of us use the terms ‘science’, ‘technology’ and ‘medicine’ without much care and attention. Imprecision is endemic, confusions commonplace. When asked to name an achievement of science, we often name a device or process that might better be called technology. In turn, the word ‘technology’, rather like ‘biology’ or ‘history’, is ambiguous; it can refer to instruments and techniques or to the *study* of these instruments. ‘Medicine’ is easier to use (if we ignore the contrast between surgery and ‘medicine’ in the sense of ‘internal medicine’). But ‘medicine’ in the wide sense sits oddly alongside ‘science’ and ‘technology’ since it includes the sciences and technologies that relate to health and disease. In that respect ‘medicine’, as an omnibus term, resembles ‘agriculture’, or maybe ‘engineering’ or ‘electronics’ – medicine can include anything from folk remedies to brain scanners, as ‘electronics’ spans from computer games to particle physics. These omnibus terms are very useful to historians partly because they avoid, or at least postpone, questions about what counts as science rather than technology. But, unfortunately, we do not have a full set of omnibus terms that can parallel ‘medicine’ – we still talk, for example, about chemical *sciences and technologies* – nor do we have a word that would encompass all these parallel science-technology fields. If we want an expression to include all of science-technology-medicine, in other ages as well as our own, I fear we must ‘develop’ one – which is why I lump science, technology and medicine as STM. This acronym does in fact have some present currency, both among librarians, and in the titles of academic departments such as mine – a Centre for the History of STM.

But that usage, and that convergence of historical studies is

new. The histories of science, and of technology, and of medicine have usually been studied separately – often in separate faculties of universities, where historians have taught (for) scientists, engineers or doctors. Partly because of this separation, the conventional chronologies of the histories have been different, and our bringing them together forces a rethinking of their outlines and major features (Kragh, 1987). Whereas histories of *science* have tended to highlight the scientific revolution of the seventeenth century, most histories of *technology* have hinged (or begun) with the Industrial Revolution c. 1750–1850. Histories of *medicine* have been less ‘revolutionary’, though the ‘birth of scientific medicine’ is usually dated about 1870, and the ‘Birth of the Clinic’ in France c. 1800 has its supporters. This book is intended to help pull all these different histories together, to allow them to illuminate each other, and so to effect a synthesis that may work across most of STM.

Indeed, my synthesis extends beyond our current definitions of STM, for I follow where my ways of knowing lead. So under natural history I include archaeological remains and collections of pictures; my treatment of analysis includes economics and language studies, and a little on the social sciences, because these were continuous with analysis in the natural sciences and medicine; and in discussing the ‘meanings’ of the world and of STM, I explore some of the philosophical and cultural issues around STM, and the layers of human values that gave life to its projects. Science, technology and medicine are the focus of the book, but it has grown beyond them. Only by looking beyond STM can we see its limitations and its importance.

Dissecting out the elements of STM

My third key is dissection or analysis. As I tried to illustrate above, I take apart science, technology and medicine into ‘elements’ which I call ‘ways of knowing’.² I focus on three ways of knowing, three ideal types of STM which I call *natural history*, *analysis* and *experimentalism*, but I also discuss a form of STM which I label *technoscience* and I explore the historical variety of

‘natural philosophies’ or *world-readings* – the *meanings* of nature and of STM.

The first three are ways of investigating nature or human artefacts. As noted, objects or systems can be described and classified (the activities I group as natural history); they can be taken apart by hands or just by brains, into their *elements* (this is the core of analysis); and the elements can be ‘rearranged’ to produce new and interesting phenomena (which will serve for now as my characterisation of experimentalism). I use the term *technoscience* primarily to refer to the production of ‘scientific commodities’ in academic-industrial-governmental complexes, such as the military-industrial complex or the medico-industrial complex. My term *world-readings* is meant to refer both to the ‘decoding’ of the world and to the systems of meaning found there. It covers the moral significance of eggs, the messages in the stars, the body as a divine creation or thermodynamics as foretelling the end of the world. It corresponds in some ways to ‘natural philosophy’, except that the world of human creations is included.

I will develop examples below, but we can note here that these ways of knowing are ‘ideal types’ in the sense of the German sociologist, Max Weber. To give a Weberian example – ‘bureaucracy’ was the rule-governed form of organisation paramount in government offices; the word could refer to a massive system of offices or to the workings of a few people, and offices could also contain other social types, e.g. friendship or charismatic leadership. Similarly, my ways of knowing can be used to characterise a multi-laboratory project (perhaps as chiefly analytical), or to describe its particular components. Many scientific projects will contain more than one kind of knowing; indeed, in my usage, analysis presumes natural history and experimentalism presumes both; technoscience projects typically include all my other ways of knowing. And by including a chapter on world-readings, I stress that all our knowings and all our creations are underpinned by systems of meaning and values.

We can go further and think of all these ways of knowing as potentially ‘nested’. For example, experimenters commonly use

analytical methods, and they also have to know the ‘natural history’ of materials on which they work. But note that nesting can occur in any sequence: relatively complex projects may have simple goals. We might regard the exploration of the moon as largely a matter of mapping or of natural history, but getting there involved lots of analysis and experimentation. Genomics – the spelling out of the human genetic code and the measurement of its variation – is best described as an analytical project, but its techniques involved much experimentation, and the collection of ranges of samples from the human gene-pool will draw on our natural-historical appreciation of the variety of humankind. Generally, STM projects are not to be ‘dropped’ into single ways of knowing, like specimens into boxes; rather, ways of knowing are to be used in analysing the *components* of projects and the relations between them. If you like, they are the *elements* of my analysis.

I will show that each of these ways of knowing has a history, and that these histories differ; new ways of knowing are created, but they rarely disappear. As Western society has grown more complex, so ways of knowing and doing have been built up. These ‘ways’ or projects interact in various ways and their ‘coverages’ vary over time. All my categories could be used, in principle, for any time and place, but as a matter of historical fact they became important at different times and in different combinations. In this view, history of STM is not a matter of successions, or the *replacements* of one kind of knowledge by another; rather it is a matter of *complex cumulation* and of simultaneous variety, contested over time, not least when new forms of knowledge partially *displace* old forms.

We will explore below the historical course of power relations between different ways of knowing, but an example here may help. I will argue in Chapters 4 and 5 that we might view the early nineteenth century as the ‘age of analysis’, by which I mean that analysis, both in my judgement and in the eyes of contemporaries, was the new, exciting, *dominant* way of knowing. But that does not mean that natural history was in decline, or that no one was doing experiments, or that nothing in that period

could usefully be described as technoscience. The contrary is the case, as we shall see; in that period, as in all others, STM was manifold.

With those preliminary comments, we can now survey the ways of knowing (and later we can link them with ways of making); at the end of this chapter we will see how these separate histories might fit into a chronological pattern. But if you are growing tired of ‘method’, skim forward towards Chapter 2.

Natural history

‘*Natural history*’ like the classical Greek ‘*historia*’, comprehends the variety of things, whether human-made or natural, ‘normal’ or ‘pathological’. I include the natural history of animals, plants and minerals, but also phenomena such as the weather, and artefacts – ancient and modern, exotic or industrial. In Chapters 2 and 3 I will discuss the creation of early modern ‘cultures of fact’ and the social history of different kinds of collecting, describing, naming and classifying; here I want to stress two points.

First, the coexistence of the two dimensions suggested by the very words ‘natural history’ – ‘natural’ suggests the taxonomic aspect, the *range* of nature (or human creation), and ‘history’ the biographical aspect of nature (or artefacts) *in time*. The taxonomic dimension is obvious enough – we shall discuss classification and displays from the Renaissance to the present day; ‘biographical’ in this context may be less evident, but it is important. Medical case histories are obvious examples of ‘biography’, and throughout this book we will refer to biographical medicine as a continuing tradition of seeing illnesses as disturbances of individual lives; but towns could also be chronicled, and so could ‘nature’ – think here of Gilbert White’s eighteenth-century classic, *The Natural History of Selborne* – his collected letters recording the changing seasons in the countryside around his English village.

Secondly, I stress the continued importance of ‘natural history’ and of ‘information’ more generally, even in the most technical of sciences and of industries. Manufacturers and experimenters

need to know the range of materials available and the often subtle distinctions between them (Stansfield, 1990); they use many catalogues and much 'experience'. We live now in an 'age of information'; we feel excited, maybe overwhelmed, by the mass of 'facts' at our disposal. That perception may perhaps animate our appreciation of classifiers around 1700 who struggled with the huge variety of 'specimens' brought back by explorers and traders.

From the seventeenth century, at least, natural history has been the study of 'what we have' – in data banks, or public or private collections, or when we take our 'complaints' to the doctor and wonder 'what we've got?'. Pride of recognition, possession and presentation is often a motive force – whether for the 'cabinets of curiosities' collected by Renaissance princes, the huge national museums built by the imperial powers of the nineteenth century, or the moss specimens collected by knowing artisans of industrial Manchester. But, additionally, as we shall explore in Chapter 3, natural history in my extended sense was and remains crucial for trade and industry.

Analysis

If natural history records variety and change, *analysis* seeks order by dissection. I will argue that analysis comes into play when objects can be viewed as compounds of 'elements', or when processes can be viewed as the 'flow' of an 'element' through a system. Analyses can be of many types, even for the same objects of study. Thus, for some purposes, we can reduce minerals, such as rock salt or sea salt, to formulae such as NaCl (sodium chloride); for other purposes we could 'reduce' them to their constituent 'unit crystals', or treat them as electrical materials and measure their conductivity. And processes can be represented by formulae, e.g. by saying that a steam engine achieves 50 per cent of its theoretical power output.

The divisions of natural history correspond to 'ranges' of natural things – plants, say, or birds – but each *analytical science* is constituted by the use of a particular kind of 'element'. Analytical chemistry reduces the whole world to chemical

elements, thermodynamics to energy, and histology is the study of tissues in whatever animals they occur. These 'elements' were not obvious; in some sense they did not exist before 1750. They were 'discovered' and corresponding disciplines were 'invented' – mostly in the several decades around 1800.

In as much as analysis changes our understandings of objects, it may involve new kinds of classification and displays – 'deeper' than those of natural history. It is intimately associated with *comparative* work because common elements give new frameworks for comparison. In Chapters 4 and 5 we shall discuss the historical connections between analysis and collections, and the technical roles of museums, observatories, teaching hospitals and 'field stations'. I shall argue that the sciences of the early nineteenth century were primarily analytical, and that the new sciences were largely created by teachers of technical professionals (chiefly engineers and doctors), and that analysis remains a crucial, if much neglected, aspect of modern science (I also include sections on the human sciences that seem to me analytical, including political economy and the study of languages). All the analytical sciences were and remain hugely important for the improvement and regulation of technical processes. Most of the 'scientific practitioners' of the nineteenth century were analysts, variously working in industry, agriculture or medicine; that may be true for the twentieth century also. (Historians of science, unlike economic historians, rarely collect statistics; we scarcely have the categories.³)

Experimentalism

If analysis is about taking things apart, *experimentalism* is about setting them up. The former is about specifying the composition of the 'known', the latter about putting together elements and controlling them to create new phenomena (or old phenomena in new ways).

My usage here is deliberately narrow. The early modern practice of collecting peculiar or impressive phenomena in 'experimental histories' I treat as part of natural history, part of cataloguing and displaying the world. 'Experimental measure-

ment' is often best seen as quantitative analysis, whether in chemistry or physics, engineering or medicine; like other kinds of analysis, it is crucial to the refinement of technical processes. Experimentalism, in my terms, builds on analysis; it is about 'synthesis' and the systematic production of novelty. Without in any way denying the importance of earlier exemplars, I present experimentalism, in Chapter 6, as a set of practices chiefly institutionalised and theorised from the mid-nineteenth century, especially in universities newly oriented towards 'research' (rather than chiefly to the education of professionals).

Synthetic chemistry is a key example. Chemists who understood the elements (and structures) of a given class of chemicals could go beyond analysis. They guessed how to make complex compounds from simple ones; in some cases, the synthesised compounds were entirely novel. Physicists, creating a tradition of experimentalism, systematically explored the 'reactions' between such 'elements' as light and current electricity. We might argue that the creation of the phenomena of radioactivity, from the 1890s, was at once an extension of reaction chemistry and of this 'reaction' physics. But the example which I use most, and on which I based this understanding of experimentalism, was drawn from neither chemistry nor physics, but from the third great area of later nineteenth-century science – physiology. In his *Introduction to the Study of Experimental Medicine* of 1865, Claude Bernard (see Bernard, 1957) used his own researches on animal functions to show the potential of experiment in medicine. The text introduced, or at least defined, the 'control experiment' familiar to generations of scientists thereafter – when one repeats the experimental set-up in all particulars except the key variable, to try to ensure that the variable is indeed responsible for the effect in question.

And when the novel products of experimentalist laboratories could be developed as industrial commodities, then *technoscientific* complexes could be created and exploited.

Technoscience

Technoscience refers to ways of making knowledge that are also ways of making commodities, or such quasi-commodities as

state-produced weapons. In as much as many governments over recent centuries have used and supported 'natural philosophers', doctors, engineers or indeed astrologers, technoscience is a 'time-less' category, a term with possible application in many societies and one that can be related in various ways to our other ways of knowing. Where state support and commercial goals were tied up with exploration and inventory, as in major national expeditions, it may be useful to think of a kind of technoscience in which *natural history* was the predominant way of knowing. Similarly, the interconnections around analysis were sometimes dense enough that we might usefully refer to *analytical* technoscience, for example in Revolutionary France or the industrial cities of early nineteenth-century Britain. But though state support was important to natural history and crucial to the formation of many analytical disciplines, I would argue that the interests of governments, of academics and of commercial companies were still substantially distinguishable from each other in the mid-nineteenth century, at least in peacetime. The technoscientific networks were still very marginal to government, to the direction of work in most universities, and to most industrial companies.

It was from *c.*1870, I will argue, that we begin to see the development of more inventive, intense and self-perpetuating synergies between the three sets of interests. Initially, of course, these technoscience networks were small; they involved natural history and analysis, but also, and crucially, they involved the systematic production of novelties by the interplay of experiment and invention. We may specify this form of technoscience as *synthetic*, and this is the formation I see as increasingly central to twentieth-century STM. The two initial networks remained crucial – one around the new electrical industries and the other around dyestuffs and pharmaceutical companies. States became involved as consumers of these scientific commodities and in some cases as producers, but also as regulators, especially for standardisation. Electrical systems, within and between nations, depended on agreed measurements and units; so did the use of new 'biological' treatments such as anti-toxins or vaccines. In as

much as the products proved saleable or useful in war and public health, so companies, universities and states all had an interest both in developing further products and in reinforcing the ‘innovation systems’.

The growth of technoscientific *companies*, producing new commodities from their own research laboratories *and* from the web of academic and governmental institutions, has been a major feature of the twentieth century. But my term ‘technoscience’ is also meant to cover high-tech projects funded and directed by state agencies, or even by state-backed academic consortia. Projects such as the US space telescope or the high-energy accelerator run by CERN (Conseil européen pour la recherche nucléaire – European Organisation for Nuclear Research) at Geneva, could be said to involve ‘commodities’ in the extended sense of products wanted by governments; they certainly involve dense intertwinings of universities, industry and government. For all such systems, trying to separate the science from the technology seems less profitable than recognising their characteristic, dynamic combinations. They all involve many kinds of natural history, analysis and experimentalism, and many kinds of making; they also involve massive organisation, and now dominate the academic and industrial worlds.

World-readings (or hermeneutics)

‘Reading the world’, by contrast, is as old as humankind. The various ‘understandings’ of nature and of STM will be explored in Chapter 2, where I sometimes refer to this decoding work as *hermeneutics* – a term that once meant the art of interpreting sacred texts, but which can be used more generally for the art of reading ‘meanings’, whether in texts, artefacts or nature. The products and frames of such decodings we might call ‘natural philosophy’ – if we extend the usual sense so as to include reflections on human creations and creativity, including the structure and processes of STM. For sure, these are not easy issues to pin down, especially at the start of a book. To discuss them, I borrow from accounts of science and religion and of science and literature. I also use work that is classed as philosophy and I draw on

some 'political' histories of science and medicine, but there are no existing histories which cover the full range that I want to bring into focus.

I will begin with natural worlds that were meant to be read as systems of symbols, starting with the courtly worlds of the Renaissance. We glimpse how scholars interrogated texts, decoded the particulars of nature and used the clues to tweak the world by 'natural magic'. From this world of natural meanings I turn to northern, more Protestant worlds to see something of the (partial) 'disenchantment' of nature. The culture here is of navigators and merchants, of urban markets and coffee houses, and of aristocratic estates and churches in the country. It was a world created by God for men and women who also created, and who stood before Him as individuals, without the intermediaries of saints, cardinals or divine monarchs. We shall explore the cultures of accumulation and of agricultural improvement that were the context for natural history and 'experiment'. We shall also see how analysis was linked to navigation and surveying, and proved fundamental to the new 'cosmologies' of mechanisms and laws.

The 'world-readings' there established were mostly about *order* and *eternal truths*, but this book is mostly about *science and doing*. For this aspect of STM, the formative period is the decades around 1800 in France, Britain and Germany. There, we might say, were invented the three roles which made much of our modern world – the *technical-professional* (new kinds of engineers and doctors), the *industrialist* and the *researcher* (cf. Ben-David, 1971). Of course, such a claim needs massive qualification and explanation, so hold on to it for now and judge it later in the book. But the simplification may help us focus on the new 'meanings' of STM since the 1800s, and their interactions with older meanings. For example, the analytical sciences which I describe in the professional schools of post-Revolutionary Paris could be used for 'rule by experts', otherwise known as technocracy and still associated with France. Our attitudes to industry – the qualified welcome for the efficiency of rationalised production and for the creativity of inventors and

industrial scientists, but also the worries over the mechanisation of people and the ‘commercialisation’ of everyday life – can be traced back to the Industrial Revolution in Britain. And the third role – the university professor – whether in physics or in the study of ancient languages, is rooted in the reform of German universities c.1800. Professors still inhabit our public worlds, though now we sometimes wonder by whom they are paid.

But just as the STM of the early modern period cannot be discussed without theology, so that of the modern period also requires attention to wider cultural frames – especially to ‘readings’ of human life in philosophy, literature and history, as well as in theology. The period around 1800 produced professors of philosophy and of literature, as well as of natural sciences. Even more important was the construction of a fourth new role – *the creative, individual, ‘romantic’ artist or writer* who taught new ways of finding meanings in nature and in human history. Romanticism will also be part of our discussion in Chapter 2. Like other ways of deriving meanings, and like other ways of knowing, the role of the artist can in principle be understood as a means of intervening in the world.

Ways of knowing as work

The fourth key to my history of STM is ‘work’. I have tried to think of ways of knowing not just as mental operations but as modes of *work* – e.g., the work of analysing chemicals or pathological specimens, with all their characteristic locations, requirements, routines and products. In this respect my approach accords with much recent research in history of STM which you can find in specialist journals, but it is distanced from most of the general histories of science, even those written recently. These ‘surveys’ tend to focus on theories and ideas, such as ‘evolution’ or ‘energy’, and/or on the histories of particular disciplines, such as astronomy or chemistry, rather than on the forms of practice which may extend across (and help constitute) many disciplines.

By concentrating on the work of science, I hope to bring history of STM closer to social and economic history, and to

facilitate studies of the kinds of scientific work that were *concurrent* at any given time. For example, we could in principle ask how many ‘analysts’ were at work in the universities and industries of late nineteenth-century London or Boston, or how the labour forces of natural history grew over the century. Such histories would be quantitative, like some studies of industrial history; more generally, they would be about synchronicity and interdependencies – not just about novelties and new frontiers (Edgerton, 1999). Though, for the most part, such histories lie in the future, thinking in these terms is useful now, not least to deepen our understandings of change.

Scientific change involves *changes of work patterns*, not just of ideas, so the *institutions* of STM are a key part of this book. For example, the emergence of ‘clinical examinations and pathological anatomy’ in Paris after the French Revolution, was much more than a theoretical shift in medicine; it involved major changes in professional and educational structures, in the control of hospitals and in the work routines of doctors. It could not have been realised without the political changes and, had those changes been different, then *different* medical aspirations would have been put into practice. Similarly, the rise of experimentalism in physics as a regular practice and as a social institution had many and various preconditions, including advanced and systematic education, analytical concepts and practices, workshops and laboratories for the construction and use of experimental systems, plus the time and money needed for would-be experimentalists to devote themselves to these creations, rather than to the demands of, say, commercial analysis.

It was in exploring such historical relations between knowledge and practice, or science and technology, that I came to think about three ideal types of *making* which I call *craft*, *rationalised production* and *systematic invention*. At first I thought of them as a way of explaining my ways of knowing, through the analogy of knowledge production with manufacturing. For example, we can use the three ‘types’ of making to reconstruct the history of manufacturing in terms of displacements rather than replacements: the Industrial Revolution around 1800 could be seen as

based on rationalisation and mechanisation of production, while recognising that crafts continued to be important (Braverman, 1974); systematic invention, I suggest, became important at the end of the nineteenth century, when industrial research laboratories were created for that purpose. My ways of *making* functioned in similar ways as my ways of *knowing*, and they also 'nested' similarly.

It still seems to me that the ideal types of making may be a useful tool for the history of technology – not just for the history of manufacturing, but also for agriculture, medicine and the military. Craft, rationalisation and invention can be ways not just of making, but of tending, mending, defending or, indeed, of killing – think of keeping a pet hen, of battery-farming chickens, and of genetically engineering new breeds of poultry; or of duelling, trench warfare and guided missiles (and of all their possible coexistences and interactions!). But I came to realise that the relationship between these ways of knowing and ways of making was much more than analogical.

I would now argue that there are close, systematic linkages, recognition of which helps illuminate many common questions about science–technology relations. It is not hard to see that rationalised production, which typically involves the deconstruction of craft activities and their reconstruction as machines, depends on an 'analysis' of craft work similar to that involved in the deconstruction of chemicals into their elements. And I have probably already said enough about my perspective on 'experimentalism' for you to see the close kinship with 'invention': both create novelties from elements – experiments are usually for enlightenment, inventions usually for profit. Indeed, experiment and invention, like analysis and rationalisation, might be viewed as two sides of a single coin. But what of craft and natural history? – not a combination much discussed by historians.

It would be trite to claim that all natural history involves craft, though it is *not* trite to stress that all ways of knowing involve crafts (Polanyi, 1958) and several recent sociologists of science have stressed the role of tacit knowledge and skills, even in mathematics (e.g., Collins, 1992). It is more specifically enlightening

to note that most early modern (and many late modern) technologies, e.g. agriculture, textiles and medicine, involved much special knowledge of minerals, plants and animals, as well as craft skills. Although these characteristic alliances have been hidden by historians who concentrated on industrial technologies, they are coming to light in newer accounts of the 'scientific revolution'. We note, for example, that most museums contained both the works of God and the works of humans. But we can go further. Craft products, I will argue, were commonly seen as quasi-species in much the same way as animal and plant species. Think of the *local* varieties of cloth – the 'blue denim' of 'jeans' was once 'Bleu de Nîmes', a variety of cloth made near Nîmes in the South of France, with characteristics peculiar to the place; or of wines such as 'Bordeaux' or 'Champagne', and the ways in which they depended on (and helped constitute) both region and season. Vernacular housing or traditional kinds of spades show much the same pattern, and like cloths and wines they could suffer from characteristic defects and 'diseases', as we shall explore in Chapters 3 and 4.

Thus, summarising very tightly the relation of knowing and making, we can see both 'extended natural history' and crafts as aspects of material culture – including the creations both of people and of nature; they concern the 'thinginess' of life – gathering and making, possessing and displaying. Analysis and rationalisation are employed both to 'deepen' classifications and diagnoses, and to refine and regulate technical processes, especially in medicine, agriculture and industry. Experiment and invention are aspects of the search for novelty and the creation of model worlds; they can be commercialised or put to state uses by technoscientific networks that manufacture 'knowledge commodities'.

My setting out these claims about ways of knowing and ways of making is itself a novel enterprise. It may seem strange to readers at this point; indeed I feel myself that the historical project is only just begun. But I hope thereby to open up new avenues by which our histories of science, technology and of medicine can be combined in new and fertile ways.

Missions for this book

If we take together my four keys – breadth, history, dissection and work – we begin to see the project of this book – to present the STM as a variety of changing projects within the economic, political and cultural history of the last few centuries. By analysing STM we can show how its constituents are part of those wider histories and thus part of our varied heritages.

Ideally, I would include other traditions than the Western, but this I must leave to others. I concentrate on Europe and North America over the last 300 years, and especially on Britain and France, but even within that frame the coverage may be thought very uneven and excessively Anglocentric. I am aware that like so much history of STM, my account says relatively little about military STM. I have missed out much that I could have included, for this is an introductory text not a comprehensive survey, yet, even so, I have ventured into many areas outside my special fields of study, and doubtless I have made mistakes. The text is exploratory and provisional – introducing a method and trying it out by raiding, rearranging and sometimes revising the works of my fellow historians. I shall be pleased to receive corrections and suggestions.

In many cases I have used local, even personal, histories. I have often drawn my examples from Paris and Manchester, the places I know best. The importance of those two cities in the Age of Revolutions (French and Industrial) will be evident; so, I trust, will the virtues of comparative history; but I have a further reason for concentrating on them. By selecting different examples from the same places, for different times and topics, I hope to give the reader some feel for the ‘local connectedness’ of developments in STM. Indeed, I would like to think that readers will be encouraged to think about that connectedness in the cities and countries about which they know and care. I love local history; I have written a very general book, in part to facilitate such studies.

I sometimes use personal examples for similar reasons – to encourage historical reflection on how STM ‘plays out’ for each

of us now, and to show how the work of academic historians of science can itself be understood historically. The final chapter is unashamedly local, in time and space. Even more than the rest of the book it draws on British material, and it clearly arises from the public politics of STM around the year 2000. I am told it will date quickly and I tend to agree. But even when dated, it may still serve as an example of how history may be used, and why it is created. I very much wish that the later twentieth-century sections of this study formed a better bridge between historical recreation and reflection; but even so, it seems worth making the links explicit in the hope of increasing the utility (if not the consistency) of the whole book.

Maybe it is also appropriate to spell out here the audiences for whom I hope the book will be useful. I envisage three overlapping groups:

- 1 historians and other analysts of social and intellectual change;
- 2 scientists, engineers and doctors, and all the other women and men who are professionally concerned with the increase and use of expert knowledges and techniques; and
- 3 the general reader – or perhaps the general reader in all of us – the citizens and consumers, patients and politicians, merchants of change and champions of conservation who seek better understandings of the roles of expertise in our world.

It is not easy to address the needs of all these groups simultaneously, so please feel free to skip those of the next three sections which concern you least. If you have never heard of the historians I am about to discuss and/or you are not professionally concerned with STM, do not worry – the book does not require such knowledge, so fast forward now to ‘For the general public’, and ignore all the bracketed references in the text.

For historians

For historians, especially for researchers and students in the history of STM, I hope this book will serve as a tool kit and a new map. It draws on recent studies which are valued in these

communities of scholarship, and no doubt it will be judged as to how it 'fits' with them. I like to think it will enrich the best of them, enhancing them by comparisons and contrasts – by an oblique light. Like all works of (analysis and) synthesis, it depends on the work of hundreds of scholars. History of STM, in its various guises, has thrived in universities since the 1960s, especially in the USA and Britain; this book is possible only because of that innovative (and collegial) scholarship, and some of my debts are recorded in the Bibliography. But this is not the place for the detailed analysis of recent work,⁴ rather I now introduce some of the authors now dead or long retired who provided intellectual sustenance and inspiration: Michel Foucault, Thomas Kuhn, Lewis Mumford, R. G. Collingwood, Owsei Temkin and Max Weber – major figures all, or representative, but in very different ways and rarely for the same audiences. The texts I have found most useful are listed in the Bibliography;⁵ here I give a few notes on how I have used these authors.

Foucault is a mainstay for medical and cultural historians, Kuhn for historians and sociologists of physical sciences. I use Foucault chiefly for his analysis of 'the archaeology of knowledge', for his explorations of the different 'epistemes' (or ways of knowing) which he presented as fundamental to each period of the modern world. Kuhn is best known for his idea of 'paradigms' – models of process and outcome which can be followed by scientists in any particular discipline so as to gain (and recognise) correct results;⁶ pursuit of paradigms constituted 'normal' science; scientific 'revolutions' were prompted by the breakdown of paradigms. These ideas are helpful, especially for the period around 1800, but in this book I draw less on Kuhn's *The Structure of Scientific Revolutions* (1962) than on an article where he tried to map out the history of STM, from physics to biology, from the seventeenth century to the nineteenth (in Kuhn, 1977). His distinction there between empirical and mathematical traditions is one of the roots of my distinction between natural history and analysis.

But neither Foucault nor Kuhn has much to say about technology; here my favourite 'older' author is Lewis Mumford, a

disciple of the biologist and town planner, Patrick Geddes, and a representative of a critical history of technology first developed between the World Wars, drawing inspiration from that most astute historian of nineteenth-century technology (and Manchester) – Karl Marx. From the inter-war period too, comes the modern tradition of history of medicine, still represented by Owsei Temkin – developed in Germany, transplanted to the USA in the 1930s, submerged somewhat in the positivistic 1950s, but re-emergent from the 1960s as a new generation sought to assess the relations between medicine, culture and society. It was an émigré contemporary of Temkin, Erwin Ackerknecht, who introduced four ‘types’ of medicine seen in history – book medicine, bedside medicine, hospital medicine and laboratory medicine (Ackerknecht, 1967), which were later developed and popularised by Jewson (1974; 1976). These served as my starting point for the historical analysis of science and technology more generally.

My debts to Collingwood and Weber are more distant and diffuse. To Collingwood, for the tradition of history as understanding which he represented almost single-handedly in Britain between the wars; for his discussions of magic, craft and art which seem to me much more useful than their present obscurity would suggest; and for his childhood reading of antique science books by Lake Coniston, from which he learned that science was made like other human activities – over time. To Weber, albeit indirectly, I owe the chief method of this book – the use of ‘ideal types’. I use ideal-typical ‘ways of knowing’ in the way that Weber used ‘bureaucracy’ as a type of authority – and indeed as a way of knowing and a way of working. It may seem odd to make the acknowledgement here: the method is widespread, it did not originate with Weber and I have not studied his writings in detail. But he stands as a model for his range and his clarity; and we need those qualities as targets, however distant (Weber, 1949).

By using Weber’s method I have tried to create a typology of knowing (and making) which can be used across the whole of STM to explore concurrent patterns of work. I will argue that we can characterise periods by the relative importance of differ-

ent ways of knowing, and I will attempt explanations of changes *between* periods. But I also want to stress the ‘longue durée’ histories of various ways of knowing; revolutionary changes in science may *displace* previous ways of knowing, but they do not wholly *replace* them. Between periods, places and disciplines, my method is analytical and comparative – finding elements in common, showing how they developed and how they were differently arranged and structured. At the end of this introductory chapter I will sketch the outline history of STM which seems to me to arise from this method of analysis, showing how the usual ‘periods’ may be characterised, and how we may trace patterns of succession in ‘newly dominant’ ways of knowing. But it is central to this book that science, technology and medicine *are and have always been* much more *plural* than most people know; at any time there have been lots of different ways of knowing and of making.

This plurality also applies to *historical* studies, in the past and now. There are lots of ways of doing history, because it serves many purposes and many audiences. (There may be lots of ways of using my typology to produce outline histories different from that presented here.) Yet professional historical studies, like the various components of STM, have their own rules – not ‘anything goes’, not all comparisons make sense, not all views of the past will stand the tests of evidence which historians use, any more than all theories or inventions withstood the tests of analysis, experiment or use. Critical pluralism (rather than relativism) is central to my account of STM and to this practice of its history. I hope this book will serve many kinds of historian in critically developing their own approaches.

For practitioners of STM

For practitioners of STM I hope to provide an aid to reflection and action – a means of situating their own concerns and problems within a large historical and contemporary frame, which can also include such other ‘cases’ as they know from experience, anecdote, reportage or historical writings.

Suppose, for example, we ask what a biologist might draw from the excellent biographical studies now available on Charles Darwin, or of Lamarck and his evolutionary theories *c.*1800 (Browne, 1995; Corsi, 1988; Desmond and Moore, 1992; Jordanova, 1984). Good historians no longer see such figures as simply 'contributors' to a supposedly continuous history of speculation and theory about animal evolution. They stress context, and the way that scientific problems were 'of their time'. But how many biologists, or general readers, could use such accounts in constructing a history of science in which they themselves also had a place? Scientists are still taught to think of science as accumulated truths, and to measure past scientists thereby. Though the biographies elucidate the particular connections between science and 'period', such links will remain anomalous for much of the readership. Part of my aim here is to make *many* such links, to give general characterisations of periods, linking social and political histories with the then 'dominant' and 'recessive' ways of knowing, so that particular developments can be 'placed', both as types and in a general historical overview. By using a simple typology and advancing a new, diagrammatic model of the history of STM, I hope to create accessible frameworks in which 'cases', lives, institutions and disciplines – in all their richness – can be more easily related to each other and to wider histories. The method is simple but not, I hope, constraining. Any single 'case' can be elaborated indefinitely. I do not wish to reduce the complexity of history, but to offer maps and methods of mapping that can be used at various scales – from bird's eye views to close-ups.

These frameworks should also be useful for that part of history which is not yet the past – for the analysis of our own times. How are different *kinds* of scientific work related to each other and to other kinds of work in our present? How might we think about the role of analysts or of 'information gatherers'? What kinds of intellectual and practical infrastructure are required by various sciences and technologies? Are new ways of knowing emerging with new technologies? Does it make sense to talk about 'the scientific method'? What is gained or lost when

researchers refer to themselves as ‘scientists’? (Could it be that now the effect is rather as if office workers took to referring to themselves as bureaucrats and organising national and international associations for the advancement of bureaucracy?) I particularly hope that the book will appeal to doctors, engineers and others who are sometimes described as ‘applying’ science.

In this text the notion of ‘applied science’ will appear only as an actor’s category – a term used by the people we are studying. Though I acknowledge that many people have used the phrase and acted on its message, I think there are better ways of analysing these concerns. I suspect that in Britain and maybe elsewhere, clinical and engineering investigations have been handicapped over the last century by having to operate with a model of science that did not fit *their* customary kinds of research. Doctors and engineers often focus on analytical studies rather than ‘laboratory’ experiments, but if funding committees equate science with laboratory science, and especially with experimentation in a narrow sense, then research ‘nearer to practice’ may be marginalised. We may hope that a wider view of ‘science’ may help promote a more varied research culture.

For the general public

Since the 1980s British scientific establishments have been much concerned with the ‘public understanding of science’; most of the consequent activity could reasonably be described as public relations (PR) on behalf of scientists. But it has become clear, even to the PR agents for science, that the public is not to be thought of as vessels more or less full of science; it is more profitable and respectful to see our fellow citizens as agents pursuing their own projects, with various resources and constraints. Here is a theme I want to develop in this book. Indeed, I would count it a success if it increases people’s ability to relate esoteric knowledges and technical products to the lives they wish to lead, individually and collectively.

I hope to enhance the public utility of history of STM as a means to these private and public ends – to support the roles of

citizens whether as ‘consumers’ of STM or as its ‘backers’, giving to research charities or paying tax to support universities and other state-funded research. Of course, these two sets of roles are linked, but they are worth distinguishing for analytical purposes, not least because of the growing tendency to see citizens simply as consumers (not just of market goods, but of state services and even of political policies). Remember here the sadly surprising response of a national art gallery director when asked about entrance charges: ‘the public come in free because they own these paintings’.

In as much as we are all ‘supporters’ of STM, we need to focus on the choices available to us as individuals, and via collective political machinery. To do so, we need estimates of the plurality of STM, of the possibilities of emphasising some aspects rather than others and of the potential efficacy of public policies related to STM. Generally, we need to fit the meanings of STM into our individual and collective views of politics.

History can help that enterprise by providing both tools and frameworks. I have already introduced my tools – my ways of knowing and my ways of making; I end this rather long introduction by sketching the framework. Each of my main chapters is devoted to the history of a particular way of knowing, such as natural history or experimentalism; the book is ‘chronological’ only *within* the chapters, so I here outline the overall history that emerges from my use of this mode of analysis, extending the sketches I gave earlier in this chapter. This quick survey is meant as a preliminary orientation. It may seem rather too dense for digestion, especially if you are new to the subject, but it may prove useful for reference as you make your way through the book.

An outline of the story

The coverage starts, at least in the early chapters, with the Renaissance of the sixteenth century. I would argue that its astronomy can best be seen as an analytical science, inherited

from the Greeks and pursued by mathematicians; its natural history was part of the Aristotelian tradition of the universities, but usually subordinate to a natural philosophy which saw the world as a purposive organism. More generally, and especially in the tradition named Neoplatonic, the world was read as if a text – it was full of hidden meanings – a drama; for example, monstrous births were portents.

Histories of science written after the Second World War usually present the ‘scientific revolution of the seventeenth century’ as *the origin* of modern science – as the birth of our mechanical cosmology and of ‘the scientific method’. This book is much more pluralist – it is about many contested developments at many times, and about many methods. In my account, following Kuhn,⁷ the seventeenth century saw two complex and related shifts – the (partial) ‘disenchantment’ of nature and the (partial) restructuring of the mathematical sciences.

Especially in the Protestant cosmologies of the seventeenth century, God and humankind were alike *distanced* from nature. God had created the world and was not bound by it; humans studied God’s creation – for its fullness and its regularities, no longer for particular messages. Natural-historical studies acquired a new status independent of philosophy; and human craft-creations could serve as models of the nature created by God; and nature, like man’s creations, could usefully be interrogated by experimental manipulation. My account of the seventeenth century, like recent studies of Holland and Britain, features the workaday worlds of medicine and crafts, catalogues and recipes, exploration and collections (Jardine, 1999).

Such accounts tend to marginalise what was once seen as most ‘revolutionary’ – the developments from Copernicus to Newton. These I sketch primarily as revisions of classical analytical sciences, notably astronomy – which were then combined with the new analytical mechanics into the great synthesis of the *Principia*. These developments were fundamentally mathematical, beyond the comprehension of most educated men and women, even of the savants or medical graduates who promoted natural history and looked for new ‘experimental’ phenomena.

They were, however, associated with more general and accessible shifts in the natural philosophy of such savants: crudely, the Greek world-as-organism was partially displaced by the world-as-machine. That new model fed the appetite for analysis; savants tried to understand chemical reactions, animals and societies in mechanical terms, but with limited success.

The knowledges of the eighteenth century can thus be seen as in part analytical – the ‘exact sciences’ that were more or less reduced to mathematics and mechanics, plus the more or less speculative natural philosophies which imagined how the rest of nature *might* be ‘mechanised’. But for the most part, accounts of nature were in fact natural history – both taxonomic and ‘historical’. Naturalists and physicians, savants and antiquarians described their surroundings, possessions and ailments, and followed them over time in local chronicles. The eighteenth century was the great age of classification – not just as a tool, but as a model for all knowledge. Indeed it was partly in seeking deeper means of classifying that the new analytical sciences were created from the end of that century.

I shall argue that these new achievements of analysis owed much to the entry of professional teachers into state museums, teaching hospitals and engineering schools, especially in France. In such professional contexts, in the industrial cities of Britain and in the reformed, ‘research’-oriented universities of Germany, new analytical disciplines were created. They were crucial for the refinement of technical practices; that was major reason for their invention and for their flourishing in an increasingly industrial world.

I then explore the creation of systematic experimentation around the mid-nineteenth century – on the back of analysis, but not initially in the professional training institutions or museums. Experiment was about making and displaying new worlds, it was somewhat distanced for professional practice. It was institutionalised in more theatrical sites such as the Royal Institution in London, and it was systematically built up in universities with an ideology of ‘research’, especially in Germany. I will argue that such experimentalism was often presented as fitting for industry,

as education for 'changing the world'. From about the same time, its more worldly twin – systematic invention – was pioneered in the USA and in Germany. Together they came to be exploited as (industrial, synthetic) technoscience.

In this book, the twentieth century appears chiefly in terms of developments of nineteenth-century patterns, especially the spread of technoscientific interactions. I discuss the industrial-academic-governmental complexes which produced electrotechnics, electronics and informatics, and the growth of the pharmaceutical industries. I outline some of the wartime developments, and I survey the growth of state-sponsored STM after the Second World War, including the military-industrial and medical-industrial complexes. Finally, I note the late century shifts in the driving forces of technosciences – away from state power and towards commerce.

In the final chapter I attempt to use my historical analysis to reflect on the dynamics and problems of STM around the year 2000. Drawing heavily on British material, I explore the commercialisation of science and related issues of 'public understanding'. I try to use my typology and history to illuminate the current politics of STM, and I ask about the *meanings* of STM, of our 'nature' and of our creations. Thus at the end of the book we shall return to the issues we take up now in Chapter 2 – on *readings of the world*.

Notes

- 1 For the contrary view see Ashplant and Wilson (1988) and Wilson and Ashplant (1988); for an expansion of my argument see Pickstone (1995).
- 2 The method was first presented in Pickstone (1993a) and elaborated in various ways in Pickstone (1993b; 1994a; 1994b; 1996; 1997).
- 3 But see Price (1965).
- 4 For a recent introduction and further references see Golinski (1998).
- 5 As guides to these authors (on some of whom the literatures are

enormous) see, for Foucault, the biography by Eribon (1991) and the excellent primer on his early works by Gutting (1989); on Kuhn, Barnes (1974; 1982); on Mumford, Miller (1992); on Collingwood, his autobiography (Collingwood, 1939) and Krausz (1972); on Temkin, the Introduction to his selected papers (Temkin, 1977); and on Weber, the biography by Bendix (1960), Hughes (1974) and many collections and commentaries.

- 6 As we shall note below, the analytical sciences I describe from the period around 1800 provide most of the good examples of Kuhnian paradigms – e.g. Lavoisier’s chemistry. I owe to Andrew Warwick the suggestion that the conditions for the creation of these analytical disciplines, e.g. in professional schools, were the conditions Kuhn evoked, rather unhistorically, for paradigms generally. And indeed, that the period around 1800 was the first in which such conditions could be found. In this view, only thereafter could any science be ‘normal’ in Kuhn’s sense.
- 7 Here and elsewhere my debt is less to Kuhn’s widely known work on paradigms and scientific revolutions (Kuhn, 1962) than to his exploratory essay on ‘Mathematical versus experimental traditions’ in his *The Essential Tension* (Kuhn, 1977).

2

World-readings: the meanings of nature and of science

THIS CHAPTER IS PRIMARILY a historical review of the ways in which ‘nature’ and ‘science’ have been understood in the West, but I begin by underlining the *range* of such understandings in the present. Two anecdotes about understanding ‘ourselves’ may serve to introduce these themes.

A German historian has published a wonderful book on eighteenth-century medicine, based on case notes left by a country doctor. She shows how flows of fluids – of blood, milk, sweat and urine – permeated the imagery of the doctor and of his patients. How different, she remarks, from our own times when patients go to doctors to discuss the possibility of a myocardial infarction (Duden, 1991). Well, maybe in Manhattan! But where I come from they still have pains and tensions, weakness and ‘the runs’, even if they know some pathology.¹

A former colleague is an expert in Chinese medicine and its history. Since university rules required that the students’ essays and examinations be ‘second marked’, and since we had no other expert on Chinese medicine, I was the second marker. The students’ accounts of Chinese medicine were predictably rich, but I was bothered by their accounts of Western medicine that served, implicitly or explicitly, as foil or point of contrast. The Chinese were allowed to worry about balance and imbalance; *we* were supposed to think in the pathologies of modern medical science, and as good Cartesians we were supposed to separate mind from body. Sometimes, I wondered whether the description of the Chinese approach was more accurate for ordinary Western patients, even *c.*2000, than was the account based on ‘scientific medicine’.²

These instances are not unusual. When historians and/or anthropologists examine 'other cultures' they inevitably rely on an explicit or implicit understanding of the 'home' culture shared by author and readers – and these accounts 'of us' are often simplistic in the extreme, far cruder and less pluralist than the accounts of 'the other'. This seems to me a serious problem. Partly it arises from the feeling that we know our own culture just by living in it – and so, in a sense, we do; but if we are not careful, the richness of that tacit understanding can be lost when comparisons are made explicit. Then we may adopt Western 'self-images' devised to *contrast* with other cultures (other peoples, other times), rather than to facilitate sympathetic comparison. We may equate the self-understandings of Western patients, say, with the workplace methods of (some) professional health workers, forgetting the huge *variety* of understandings which can be found in any modern hospital. One purpose of this book is to facilitate appreciation of that variety. That is why I have introduced the method of ideal types so as to point up the different ways of knowing, their histories and their relationships. We can carry forward our exploration of this method by surveying and dissecting our own, everyday, shared knowledges. Medicine, here as elsewhere, serves as a good model for other aspects of STM.

Variety in modern Western medicine

At this point, you could try an exercise: jot down a few ways in which you might think of a sore throat. Then enlarge the frame to include the ways in which doctors, or scientists or pharmaceutical companies might think of it.

It could be the result of talking too much, or the consequence of going out last night to a smoky café or walking home in the cold. It could be a family weakness inherited from your mother. Maybe you always get a sore throat at this time of year. No doubt it will clear up in a few days if you take care, keep warm and take lemon and glycerine or Scotch whisky. But suppose it does not

clear up as expected. You may become worried about more serious conditions that run in your family, or about your failure to attend to warning signs. You may feel that the prolonged affliction is a judgement on your behaviour, or a reflection of some deeper problems in your life. Maybe you need to get yourself sorted out, maybe a therapist could help. It is increasingly obvious, even in Western societies, that not all health-related expertise is to be classed as medical.

Or you could go to your doctor? He or she can tell you what infections are 'going around' and suggest the type of infection you may have, and its likely outcome. But maybe the doctor will also take a swab test and send the sample to the laboratory in the local hospital; it will report whether or not your sore throat is due to streptococcal bacteria. If it is, or maybe in any case, you could be prescribed antibiotics, to be collected from the local pharmacist. (And if the infection were a rare or serious one, the doctor would send data to the Public Health Department.) The pharmacist may be the same person who sold you the lemon and glycerine. Indeed, the shop displays a whole range of commercial remedies for sore throats, but behind the screen there is another range of 'ethical' preparations which can only be obtained on prescription. You cannot help thinking how sore throats are a boon to pharmacists and pharmaceutical manufacturers, but you also get to wondering about the difference to your prospects which you owe to the discovery of antibiotics. Where did they come from?

Outside town there is a large laboratory belonging to a pharmaceutical company. They still do research on new antibiotics, not least because bacteria are becoming resistant to the older forms. They synthesise new chemicals and try them out on bacteria on plates of jelly, or by using laboratory animals – the 'beasts of burden' that carry infections for humans and spare them from the initial trials. If a new antibiotic works well, clinical trials will be run, often in a teaching hospital staffed by expert physicians, working closely with university departments of medical science. The university scientists are looking for discoveries that will lead to publications and academic reputation; the

doctors may be credited with a new cure. The pharmaceutical managers hope to make a profit, and to that end will employ a sales force to market the drug in the medical press and by visits and 'promotions' in hospitals and general practitioners' surgeries.

In our complex society, your sore throat has many meanings: a judgement or a sign of misdoing, an instance of a disease species or a kind of inflammation, a possible public hazard, a product of a bacterium, a process that has been imitated and studied in laboratory animals, and an opportunity for the makers of medicines corresponding to one or more of these understandings. Suppose we ask about the history of this complex of understandings. Usually we will be told the 'scientific' history, roughly as follows.

Louis Pasteur, a French chemist, argued in the 1860s that fermentation and putrefaction (thought to be akin to 'fevers') were caused by simple microscopic plants – let's call them microbes. By the 1880s, these microbes could be 'cultured' on nutrient jellies, and so identified and reliably reproduced. By culturing microbes from animals and introducing them into other animals, it was shown that several key infectious diseases were 'caused by' specific kinds of microbes. Microbes could be attacked by antiseptics, if they were 'external to the body', but antiseptics damaged the body's own tissues, and so medical scientists searched for 'magic bullets' which could kill bacteria inside the body. Pharmaceutical companies invested heavily, and by the 1930s agents were known which would cure septicaemia, etc.

It was also known that some fungi could inhibit bacteria in cultures. During the Second World War a team of researchers in Oxford took up one of these fungi and showed that a stable agent could be extracted (penicillin) which was remarkably effective when injected into humans suffering from common bacterial infections. Pharmaceutical companies (first in the USA) worked out how to produce penicillin in bulk, and also invested in searches for other fungi with similar properties. By the late 1950s, several such agents were in common use, and all common bacterial conditions (including tuberculosis) could be cured. But by the end of the century it had become clear that bacteria were

not so easily vanquished; new strains had emerged which were resistant to most antibiotics. Diseases came to be seen as a matter of 'balance' between the adaptive powers of microbes and of humans.

But why give the history only for 'scientific medicine'; don't the other views have histories, or at least, distributions? What can we say about the histories of meaning, or about the histories of 'sore throats' as an official category or as 'category of nature', and one that might vary between countries (as the significance of the liver varies between European countries)? Most cultures know how to read diseases for 'meanings' and many cultures have ways of treating illnesses as 'matters of fact', akin to plants and animals in having their own characteristic forms and durations. It is tempting to assume that these forms of understanding are less important to us because they are more widely shared, or that they are less bound up with Western history because they are also part of other histories. But that may be the reverse of the truth.

Much of this book will be about the history of these simpler understandings: about *natural history* in a very general sense, and about meanings. Natural history – the arts of describing and classifying which underlie all our more complex forms of 'naturalistic' knowing – will be the subject of the next chapter. The rest of this chapter is primarily about '*readings of the world*' or hermeneutics (Ferraris, 1996), about the meanings which have attached to 'disease' and 'nature' and also to the processes of 'science', in other times and places.

Meanings and readings

Most cultures exhibit symbolic understandings. It is a commonplace of medical sociology and medical anthropology that our culture is no exception. People ask 'why me?' 'Why this sore throat?' These symbolic understandings involve 'readings' of diseases as *message bearing*, they involve decoding as if diseases were texts, and they involve the supposition that the diseases are

part of a 'meaningful' cosmos (Cornwell, 1984; Helman, 1986). There is no originality in my pointing to these aspects of our culture, but originality is not the only end of scholarship, and some commonplaces bear repetition, even emphasis, because they are often 'missed out' of standard accounts. Especially in the history of science and technology we need to remember that understanding 'things' as things is *not* the basic currency of our individual and collective lives. Such 'naturalistic' understandings are very important aspects of culture, and they have been remarkably elaborated in Western cultures, especially over the nineteenth and twentieth centuries; but, at root, we live in a world of 'agents', meanings and feelings, and of communications (face to face, distant, or imaginary).

In this world of purposeful actions and meanings we may include non-humans, sometimes in ways we now class as allegorical or metaphorical rather than 'real'; we may talk to trees or learn to trust a climbing rope. If asked to explain such 'communing', we now resort to pop psychology. We would excuse our talking to trees as a means of focusing our own spirits rather than theirs. Or, rather, some of us would, for some purposes – when we are operating in modern-Western-knowledge mode. But where did that mode come from? Whence the divide between *our* minded bodies and the trees' mere wood? Why is the 'meaning' of my sore throat a matter for my friends or therapist, rather than my doctor? Those divisions have histories.

As we see so clearly now in debates about birth, death or animal rights (and eggs), individuals and cultures vary in the boundaries they sketch between 'agents' and 'mere objects'. But whenever we *contrast* Western 'naturalism' with non-Western 'symbolic understandings', we get off on the wrong foot and introduce a basic asymmetry that is unhelpful, both intellectually and ethically. For even when we deal with the most 'modern' aspects of Western culture, 'readings' and hermeneutics must be to the fore. How else shall we understand modern commodity cultures and their advertising? It is no accident that public relations firms employ anthropologists; and if magic is the technology of meanings, or the art of raising or depressing morale (as

Collingwood, 1938, maintained), then are not advertisers, PR merchants and ‘spin doctors’ its prime contemporary practitioners?³

That is our world now, but it derives from a series of different worlds over centuries. This chapter ranges from the Renaissance to the present, from theological contexts to political, and from the cultural roles of doctors to those of poets. It is intended as a ‘background’ for the more detailed chapters which follow, and as a way of placing technical developments in wider contexts. It is also a preparation for the final chapter, in which I will discuss the meaning of STM in our times. Some readers may therefore wish to return to it after the rest of the book but before that final chapter.

Let us turn (in)to the magicians of the Renaissance.

Renaissance cosmologies

In some of the literate worlds of Renaissance Europe, it was customary to be joined in meaning with one’s surroundings. The world had been created as meaning-full; plants that were good for curing eyes announced the fact by their colours and design. Yet some messages were hidden, and expertise was needed to break the code. The (short) history of the world was written in books, especially the Bible; and from Arabic texts, or Latin, or ancient Greek you could recover accounts of the world that had been lost in later centuries. To read the texts, and to read nature, you needed skills in language and decoding (Debus, 1978; Pumphrey, Rossi and Slawinski 1991).

To enter that mental world, think what it might be like to wake up on a stage, surrounded by a play. How could you work out what was going on and what it all meant? Clearly, you could follow the dramatic action and try to make sense of the plot. You could study the sets and the ‘props’ for their uses and meanings. Or maybe you could find a text that corresponded to the play, and so discover more about the author and his production; perhaps you could even find ways of communicating with the

author by ritual or prayer. Those indeed were the ways of alchemy or astrology – elaborate interpretive procedures or calculations, the use of ancient texts, and spiritual preparation and communion. By making such comparisons we can try to understand the Renaissance view of the world as a drama. Reflexively and ironically, we can use parts of our own experience to find the meanings of a (lost?) world, full of clues and meanings.

To understand diseases in such a world, you need to think about ‘agency’ as well as nature. This was a world of hidden personal politics and of courtly suspicion. So if you asked your physician about your ailments or about the best day to go hunting, he might focus on blind conjunctions of cosmic forces or on the machinations of malign agents – whether human or creatures of the dark – any of which might be responsible for disturbing your humours or planting in you the seeds of disease. Against unfortunate cosmic conjunctions, you might be forewarned by astrologers; for protection against the malign, you could pray, make offerings and take precautions. Such thinking is not dead, even in the ‘advanced West’, but it has not been respectable among the highly educated since the seventeenth century, when the reputations of ‘magic’ and astrology declined steeply, for reasons we shall discuss below.

To recover that Renaissance world is easier these days than it would have been soon after the Second World War. The ‘literary turn’ in history, philosophy and social sciences has familiarised much of the reading public with perspectives which would have seemed restricted and remote in 1960. We are self-conscious now about reading written texts, and some of us have been taught to ‘read’ rituals or social interactions as if we were anthropologists. We refuse to be superior to ‘primitive mentalities’, and we try to understand sympathetically the different meanings that are read into the natural world by various cultures, as well as our own. Indeed, the hold of hermeneutics in some parts of social studies, including parts of science studies, is now so firm that we can easily feel the contrary historical problem – how does one ‘get out of’ hermeneutics? How does one stop writing only of what a bit of nature meant to whoever, and start writing directly about

objects beyond texts, presuming a language that others will share (Schneider, 1993)?

Thus we can revivify key questions about the 'scientific revolution'. How were *parts* of the Renaissance world 'disenchanted'? How were they 'stripped of meaning' so that plants and diseases, and planets and eggs could (sometimes) be *merely* natural – so that they could be described in 'plain prose' that bore no trace of cultural practices or historical derivations? Or perhaps in a universal language which would perfectly transmit the characteristics of the objects or phenomena in view. Or, to put the same question in terms of ways of knowing and of historical displacements: how was a genre of natural history created or re-created in which 'natural objects' could be constituted without their symbolic meanings, so that sometimes, and for some purposes, we could think of them merely as things? And what happened to hermeneutic ways of knowing as a result of such displacements? In the forms of writing which encompassed this new natural history, where were meanings now located?

In the following sections I follow some of these questions, from the Renaissance to the present. Perforce, I concentrate on literate elites debating the boundaries between the realms of meaning and the realms of naturalistic understandings, or between 'culture' and 'nature', but I try to use 'understandings of diseases' to link elite cultures to the everyday. Here I stress again that we all still know enchanted worlds – many groups live in worlds where God's providence still holds sway, or in 'post-modern' worlds re-enchanted by media and commerce.

Disenchantment?

In his book on *The Order of Things*, the French 'archaeologist of thought', Michel Foucault (1970), gave a striking depiction of the world of symbols to which I have crudely alluded in my discussion of the world as 'theatre'. His emphasis was on *language*; in the Renaissance, words and things were on a level. Things were texts, endowed with meaning; words were to be conjured

with, themselves full of history and mystery. Foucault's vision was powerful and has proved influential but, of course, expert scholars of the Renaissance and the seventeenth century are bound to insist that neither the Renaissance nor any other period can be fully characterised in terms of one dominant paradigm. Perhaps we generalisers can meet them halfway by shifting from replacement models in which (Foucault's) 'epistemes' or (Kuhn's) paradigms *follow* each other, towards *coexistence* models in which paradigms *run alongside* each other, with different histories and in different political relations. The dominant form can then become less dominant; the once recessive or subordinate can become dominant. That, crudely, is how we explain political change, and to my mind, the model is useful, even for intellectual history. We do not have to choose between drawing out the continuities or sharpening the revolutions.

Foucault's account focused chiefly on those Renaissance understandings that scholars call 'Neoplatonic'. Philosophers who looked back to Plato stressed symbolism and the meanings embedded in the world, and some of them spoke of the 'world soul'; their doing so was threatening both to the Catholic Church and to the newly emergent Protestant churches of northern Europe. But as soon as we say that, we introduce a politics of knowledge and we begin to move from an 'archaeology of thought' to a more complex social and political history – a move on which many historians would insist, especially if we hope to provide *explanations* of ideological restructurings.

We can further recognise that Neoplatonism was not very common, even if we restrict our attention to the literate elites and so ignore the vast majority of the population. Aristotelian traditions were much more important to the Catholic Church and to the universities, many of which were religious foundations. These traditions were not about secret meanings in nature; rather, they had been carefully constructed by medieval theologians to insert the Judaeo-Christian God into a sophisticated natural philosophy derived from the writings of Aristotle. Moreover, there was much in Aristotelian traditions that would count as 'natural history,' and much, too, that we can regard as

analytical knowledge, notably the complex astronomy of Ptolemy in which the seemingly erratic movements of the planets were revealed as compounded of various circular motions.

Yet, even as we recognise the pluralism of the culture, the distinctiveness of these naturalistic and analytical traditions, and their continuities with later natural history and analysis, we can also recognise that all these Renaissance traditions stressed recovered texts and ancient wisdoms, and that classification and causal explanations were, by later standards, *subordinated* to views of the world as a single, ordered, meaningful system (Foucault, 1970). In terms of the 'nesting' of knowledge systems, we may say that hermeneutic ways of knowing were usually dominant. Natural history, as we shall see in Chapter 3, was often read as a preparation for natural philosophy, and analytical astronomical calculations were part of the hermeneutic practice of astrology. If we take medicine as an example, Renaissance practices relied on the reading of ancient texts, and dissection was used to illustrate the texts, chiefly to show the purposefulness of the body's structure – 'the uses of the parts'. The body had meaning, revealed chiefly by refined linguistic skills. When cabinets of curiosities were collected by Renaissance princes, or physicians and academics, specimens of nature sat alongside relics and craftworks – as treasures, oddities and symbols or, if you like, as 'conversation pieces'. Wonders were to be talked about (see Findlen, 1994). Hermeneutics was the overarching way of knowing nature, within which others could nest. Man understood himself through reading his place in a meaningful universe.

Thus, in the terms of a displacement model, the 'disenchantment of the world', beloved of many historians, becomes a matter of changing boundaries and relations between different ways of knowing. As we shall see in this chapter and the next, some seventeenth-century 'subcultures', perhaps especially in Protestant countries, began to give priority to 'natural history' – to description rather than meanings. We shall explore some of the causes and consequences of that displacement, but we can note now that it did not produce only a new 'nature'; it also,

necessarily, produced new kinds of literature. Devices such as allegory – *pretending* that nature had meaning, were possible only in ‘disenchanted’ worlds; only to the extent that ‘nature’ was disenchanted, could it be *re-enchanted* by language.

C. S. Lewis, the historian of literature and apologist for Christianity, described the sixteenth-century world-reading as ‘tingling with anthropomorphic life, dancing, ceremonial, a festival, not a machine’, in which, ‘the “teeming earth” can almost literally be “pinched” with a kind of colic’ (Shakespeare’s *Henry IV*). And then, Lewis wrote, ‘Man with his new powers became rich like Midas but all he touched had now gone cold and dead. This process, slowly working, ensured during the next century the loss of the old mythical imagination: the conceit, and later the personified abstraction, takes its place’ (Lewis, 1954: 6). In this view, meaning became separated from ‘the natural’, and the reading of texts was separated from readings of the world.

Most historians would agree that we see here new boundaries that were formative for later Western cultures. But if we can see such shifts as *displacements* rather than *replacements* we make more room for pluralism and for ongoing debates and tensions around the boundaries of the natural and the hermeneutic. Two such tensions seem particularly important here. To one of them C. S. Lewis alluded when he suggested that Romanticism in the decades around 1800 tried to heal the pain of what we might call the Midas effect. We shall return to this opposition and tension towards the end of this chapter, to see how Romantics such as the poet William Wordsworth then reinvested nature with moral meanings – a reinvestment which proved fundamental for later cultures, whether religious or secular. The other tension, of course, concerns theology. God remained central to Western intellectual cultures until the twentieth century, and theology was the chief vehicle for debating the relations between nature and men. Those seventeenth-century authors who celebrated such new inventions as printing or gunpowder, found in the disenchantment of nature a new basis for theology, a ‘natural theology’ of a *God who was also a maker*. This remained a crucial tradition in British and American reflections on science from the

seventeenth century onwards, and it echoes on through our own times,⁴ even for secularists or the irreligious.

Natural theology and natural diseases

It was, in part, to glorify God as a free creator, that He was freed from his creation. The intermediaries between God and man – the angels and saints, and the demons – were stripped out by Protestant theologians as part of their attack on the old Church, and ‘man’ was left to face his maker alone, as part of the religious-politics that proclaimed ‘the priesthood of all believers’. All good Protestants were assumed capable of interpreting the Bible on their own; and just as the Bible could now be read by anyone, so also the ‘Book of Nature’ was open to all. In this way, nature became a means of demonstrating the attributes of God through the study of his creation, and religion became a major reason for studying natural history and natural philosophy. In the continuing Catholic traditions, by contrast, life’s meaning could more easily be lodged in the Church authorities, in the priests and in the persistent saints. In Chapter 3 and 4 we shall meet some of the new ‘naturalists’, and we shall flesh out some of the reasons why this new view of the world suited the aristocrats, merchants and craftsmen to whom the Protestant parsons ministered.

In part this was a matter of natural history. They were all interested in selling and owning *things*, and those interests supported the collections and the data-gathering which was sometimes directly useful, but which also helped celebrate man’s possessions *and* God’s creation. Natural history was part of material culture, but it was also moral; and in as much as it was about hierarchy, it also bolstered the social pyramid. For example, in the years after the Restoration of the English monarchy in 1660, taxonomic hierarchy became a common theme both in natural history and through the cultivation of heraldry.

By including technology, ‘natural history’ could also exhibit *human creativity*. This, too, had both a mundane and a symbolic

significance – for practical life, but also for religion and philosophy. In stressing the novel creations of man, and especially the new crafts of printing and gunpowder, the Elizabethan lawyer, Francis Bacon, had given a new twist to the perceived relation between man as knower and man as creator. For Bacon and his followers, man knew *in as much as he could create*; human power was both the evidence and the fruit of true knowledge (Pérez-Ramos, 1988; Rossi, 1957; 1970). Here, was a doctrine which raised the status of ‘ordinary life’, and of crafts which had been beneath the dignity of ancient and medieval scholars. It appealed greatly to puritans, who liked to praise their God in daily life rather than by special ceremonies. As Charles Taylor has demonstrated so well, much of the cultural repertoire of modern man can be traced to this double emergence – to the inwardness of man the knower, observing nature from the outside, and to the exaltation of ordinary life (Taylor, 1989).

But the men (and occasional women) who made or supported the new natural philosophies were also interested in what we may call ‘analytical order’ – in astronomy and mechanics, in the regularities of the world. This interest, yet again, was both mundane and symbolic. Analytical knowledges and practices were important for navigation and surveying – for mastering the divisions of space and of time, but they also helped provide legitimations for social order that were independent of Church tradition; the world was regular, its regularities could be grasped, and they were grounded in God. To seventeenth-century modernists such as the mathematician and philosopher René Descartes, the ability of man to grasp the elements of nature as matter in motion, was itself a matter of divine dispensation. He and other ‘mechanists’ found models for God’s creation in human creations, especially in clocks (Mayr, 1986); their wonderfully elaborate workings could be understood without calling on their makers, but their *existence* (and maybe their forms and varieties) was otherwise incomprehensible. So too with ‘de-souled’ nature: it required and reflected its Creator. At *that* level it evoked the divinity of order.

As Keith Thomas has argued, these shifts in theology and

metaphysics were part of the explanation for the decline in the status of witchcraft and astrology, neither of which fitted the Protestant cosmology in which humans faced an all-powerful divinity (Thomas, 1971). God was not bound in the way that astrology seemed to require, and the puritan world-reading of Man facing God left little room for 'spirits'. What then did this mean for moral responsibility and for disease-causes other than 'spirits'?

The consequences were twofold. On the one hand, health and disease were matters of natural law that could be studied by naturalists. On the other, God *could* intervene directly; so a sore throat was a natural condition but it might also be divine punishment, and an epidemic could be the judgement of God on the faithlessness of his people. The *private* forms of that perspective on the world shine through the diaries of many devout Protestants from the seventeenth century onwards – their intense concern with bodily regimen *and* with spiritual health of the soul. Such attitudes were often encouraged in the lower classes, especially in times of political instability. In Britain and the USA, they were a major root of the 'self-improvement' cultivated by nineteenth-century workers, and by the twentieth-century's 'positive thinkers':

The *public* roles of 'divine punishment' were much debated in eighteenth- and nineteenth-century theology. 'Atheists' or 'deists' stressed the mechanisms and the material of nature, so casting God out of the picture, or reducing him to an original cause – sufficient to lend respectability to their own pursuits but too rarefied to serve the purposes of the clergy. But these were not just academic debates, they affected common life. So, for example, changing ideas of health and disease can illustrate the consequences of philosophical differences, and perhaps also help us explain them. The historian Michael Macdonald has provided a nice case study, in which he showed that in early eighteenth-century Britain, the magistrates responsible for local 'law and order' came to reject the tradition that suicide was a heinous sin; they exercised compassion for the relatives by treating suicides as lunatics, and by classing lunacy as a physical disease rather

than possession by spirits. Doctors were happy to agree, and to assume authority thereby, but the pressure seems to have come from laypeople weary of the strenuous religious disputes which had led to the English Civil War. They remained suspicious of religious zeal as unsettling to society, and were relieved to treat 'enthusiasm' (which meant 'God within'), not as the commendable sign of holy possession, but as a rather regrettable lack of balance (Macdonald, 1990; 1981).

Enlightenment philosophers argued about the possibility of miracles; deists and 'liberal' Protestants decided against them. When the 'better classes' of Georgian England set up charity hospitals, though they included the services of clergy, they put the emphasis on doctors and on the physical recovery of the patients. (No place here for dying in the sight of the cross, as was still the case in the great French Catholic poorhouses that we shall visit in Chapter 5) (Risse, 1999). The French might set reason against the Church, but Englishmen saw concord. Hygiene, for example, accorded with the laws of man's nature, with the laws of society, and with the laws of God. For the enlightened, a cool divinity underwrote the rational order, both in the heavens that Sir Isaac Newton had explained for his countrymen, and in the streets around the assembly rooms that Jane Austen's novels still evoke. But even then, evangelical Christians looked for personal salvation rather than divine order, and saw God's wrath in public catastrophe. That tendency increased and became more influential after 1789 when the rulers of all Europe were unsettled by the French Revolution, and the owners of Britain were worried by the new self-consciousness of the working-classes.

Revolution, respectability and evolution

After the French Revolution and in face of the unrest of the new industrial cities, theologies hardened. As natural theology was recruited to the defence of the Creator, so individual responsibility and hellfire were evoked to preserve moral order (Ward, 1972). In 1832, for example, amid the turmoil of political

reform, the Anglican Church proclaimed that the novel and frightening cholera epidemic was a divine punishment for irreligion, though 'Liberal' theologians still preferred to stress public filth and other 'natural causes' (Morris, 1976.)

British physical 'scientists' busied themselves with analysis and called on the divinity chiefly as the ground of the universe, but the 'fittedness' of animals and plants to their surroundings (and of their surroundings to them) remained crucial for *natural history*: God was the 'great gardener' who brought species into being where they fitted best. Only political radicals were likely to dispense with this explanation of the complexities of life, and they then had to use Lamarckian theories about which the leading biologists were condescending, as we shall see in Chapter 5 (Desmond, 1989). That is why Darwin's *The Origin of Species* (1859) proved so controversial; his doctrine of evolution by natural selection gave a scientifically respectable explanation for the variety and adaptation of organisms, without invoking a creator. For many of those attached to natural theology, Darwin seemed to take the meaning out of life; and in as much as *man* himself might be a product of evolution, Darwin threatened his special status as a moral being made in God's own image. It has been plausibly argued that Darwin allowed about twenty years to elapse between his insight and his book because he was worried sick about the likely public consequences of his views (Desmond and Moore, 1992).

But by the later nineteenth century, when the threat of revolution in Britain had subsided and the middle classes felt more secure, public theology gradually softened and more accommodation was sought, both between the various religious denominations and with the newly professionalising sciences. Bishops might speak for God during epidemics, but state agencies treated individual or collective disease as primarily a matter of science rather than meaning. Public education, in the new state primary schools, or in new universities and technical colleges was non-denominational or even secular. Formal prayers and Bible readings were compulsory for children, but older students were increasingly allowed to separate public learning from their

private religious affiliation (or lack thereof). Though most scientists and doctors were Christian, a few prominent scientific educators, e.g. T. H. Huxley and John Tyndall, proclaimed themselves agnostic and battled to ensure that institutions of science were developed separately from those of faith. They drew on Darwinism as science to which religion must accommodate, and on the new sciences of electricity and bacteriology as evidence of science's contribution to human welfare.

Similar shifts in public discussion took place in the USA, especially after the Civil War of the early 1860s. New institutions of secular learning (often derived from Germany) and a new faith in 'professionalism' may have helped bind a riven country and unify a religiously diverse nation. Science was high-learning, but it was also practical and good for economic development.

Generally in the West, the new theories and technology were accepted as evidences of the progress of Christian nations, but ones which could involve readjustment of faith and public morals. For example, when introduced to the world from the 1870s, the bacteria of the French and German scientists disturbed Miss Nightingale and the leaders of the hygiene movement. The microbes seemed too arbitrary in their attacks on humans. When cleanliness had meant health and dirt meant disease, then the science of hygiene had taught morals; but the lesson weakened the more that *anyone* was at risk of bacterial invasion – as often seemed the case when late nineteenth-century medical scientists tried to distance themselves from older styles of public health, and when the lobby for more state science and more science education was establishing its secular authority (MacLeod, 1996; Turner, 1980).

Science, progress and the State

By the end of the nineteenth century, 'science' seemed a motor of 'civilisation'. For some (neo-traditional) Christians, evolution by natural selection had become a substitute for divine providence. Advanced opinion saw evolution and human history as progres-

sive – as a gradual unfolding of creation which could be assisted by a better understanding of biology or of economics. This approach could be further articulated via a naturalistic study of ‘morals’ and especially via the study of ‘moral behaviour’ in children. For just as the study of epidemics suggested how you could minimise them by improving public hygiene, so studies of psychology would reveal the human condition and suggest better ways of coexisting peacefully and constructively. Conduct might be understood and guided by new analytical science, such as psychology, craniology and sociology (Smith, 1997). Around 1900, child-rearing was the object of much science, and many white ‘progressives’ were looking to the new ‘genetics’ and to ‘eugenics’ in the hope of improving their (part of the human) race by better breeding (Kevles, 1985).

Such views were common across Europe, even in countries which had experienced rather different relationships between theology, politics and STM. In Catholic France, natural theology had always been much less important than in the UK or the USA; the usual opposition was between the traditional authority of the Church and the progressive, secular, possibly republican culture of the schoolteachers and intellectuals. Evolution was elided with the progressive view of history; the analytical sciences were a basis for the technocratic politics favoured by many scientists and partially realised after the German occupation of 1870, when France sought to catch up with the technology of her stronger neighbour (Fox and Weisz, 1980; Geison, 1984).

By the end of the nineteenth century, the new technologies of electricity and medicine were the material evidence of French progress, though France’s medical champion, Louis Pasteur, did not fit the stereotypes at all well. He was a devout Catholic, whose suspicion of scientific materialism had fuelled his experiments to show that fermentation (and disease) were due to ‘life’ rather than chemicals. But he was also a fervent nationalist and a believer in the utility of science. The agricultural and industrial technologies supported by his work – on wines and beer, and on plant, animal and human diseases – were central to the ‘modernisation’ of France and its empire. His experiments promised

action against disease; the success of immunisation confirmed the potential and made him a national hero. In the cottages of rural France, charms against spirits could now keep germs away (Latour, 1987a; Salomon-Bayet, 1986).

Much of Germany was Protestant, but the 'critical philosophy' which had been part of the university culture from the late eighteenth century saved most German intellectuals from the spiritual travail of British and American believers. 'Culture' rather than God the creator was the first guarantee of social order, and German scientists learned from the philosopher Immanuel Kant how to separate the realms of scientific knowledge from questions of religious belief – they were different ways of knowing: one was about causes, one about purposes and values. True, some mid-century radicals tried to make scientific materialism into progressive politics by claiming that (analytical) biochemistry was the sure way to understand the mind, and that politics could be part of medicine, but later nineteenth-century STM was generally more conservative. Scientific industry, scientific medicine and scientific warfare became strongly identified with the imperial state.

This was true of the new industries, which the emperor backed even when professors were stand-offish; and it was true for the new bacteriology of Robert Koch, which was closely associated with the military – in marked contrast with earlier 'public health' which had been liberal in its politics and based on British models (Evans, 1987; Johnson, 1990; McClelland, 1980). The new, continental, bacteriology was not about better drains or nutrition; it focused on specific diseases, technical fixes and regulations against contagion. It was a very analytical, bureaucratic *Prussian* science, but one which still echoes as part of our repertoire when thinking about public health. To take a recent example – in the early years of the AIDS epidemic in Britain, there were proposals for the compulsory isolation of some cases, and historians were mobilised to discuss the precedents (Evans, 1987).

Generally, by the opening of the twentieth century, European and American STM was increasingly associated with the drive for national efficiency in a world of industrial and imperial com-

petition. In some ways, the horrors of industrialised warfare between 1914 and 1918 were but the tragic epitome of this modernist mobilisation. Similar arguments might have been made for the Soviet Union under Stalin, and for the National Socialist government of Germany after 1933 which allied technocratic efficiency with atavistic politics. And as we shall see in the final chapter, critics of American science in the 1960s saw government-supported STM as bound into military-industrial and medical-industrial complexes which they found alien to the liberal politics and free markets supposedly characteristic of America. Indeed, throughout the twentieth century, with more or less reason, the threat of the technocratic state was part of the public meaning of STM. But by the end of the century, it was the expanding alliance of new STM and global capital which had come to seem most problematic.

Modernist human-natures

Very generally, the more that twentieth-century science was presented as offering new technologies, remedies and protections, and the more it was identified with 'modernisation', the looser its articulation with 'traditional' society and morality. Early twentieth-century modernists argued for lifestyles that were 'scientifically natural', based on biology, informed about sunshine, open about sex (Porter, 2000). In some ways, they naturalised the hygienic rules (and the streamlined, dust-free architecture) that could be found in mountain sanatoria for the tuberculous. Such 'progressive' lives *could* be religious; they might be 'Romantic' in their attachment to nature, as we shall see; but the guiding principle seemed to be the fulfilment of human capacities by rational means.

Looking back, the 1950s now seem the high point of our collective confidence in the ability of technoscience and of STM more generally to protect us from danger and to steadily raise the 'standard of living'. Those of us who grew up under the British National Health Service, drank our orange juice and gulped

down our cod liver oil, and if we were still puny we were taken to the school clinic in town, where a rotating 'sunray' lamp spread health around the darkened room. For a quarter-century or so, post-Second World War, antibiotics and immunisation seemed capable of subduing infections, nuclear energy of providing clean power, and 'development' and contraception of raising the Third World. Because some diseases could be stopped by antibiotics, bacteria became their one and only 'cause'; the other preconditions that had worried hygienists seemed less relevant now that a cure was to hand. For a new generation, acute diseases and medicines became less 'biographical', less a matter of morals or even of lifestyles (Bud, 1993; 1998). It was only for the emergent chronic diseases of older people, that biography mattered; for the rest, diseases could be conquered and progress seemed assured – except for the threat of nuclear war.

We are less sanguine in 2000, but still the rational progress of medicine is part of our common culture. While acknowledging the links of new infections with lifestyle, our public culture does not see disease as God-given. Though 'fundamentalists' of various kinds read the AIDS epidemic as the *expected* result of moral transgression (as Miss Nightingale would have agreed), public health agencies see an *unexpected* infection which, in the West, has afflicted groups that were atypical, but neither 'abnormal' nor immoral. In this account disease is a risk, but not a morality play (Berridge, 2000).

Yet these histories are complex and ever changing. One could say, for example, that in the 1950s homosexuality was a treatable disease; for radical counter-culturalists of the 1970s it was a nature that should be expressed; and now, perhaps, a set of inclinations which the wise will recognise in themselves and in others, but which *require* neither suppression nor expression. Yet one could also say that all those attitudes have been variously in play throughout the last half-century. We could trace the conflicts between their proponents, and consider the ways in which the 'balances of power' depend on formal politics, and how they vary between nations and regions.

But in doing so, we would have to recognise that we are not

the only historians; the people we study see themselves in history. They present themselves as progressives, or as custodians of traditions which have withstood repeated attacks; they draw on formal and informal histories to constitute themselves and their campaigns. More than 'nature's laws' are at stake in determining appropriate responses, and even if we focus on the interpretations of nature, modern STM is not the only authority. Present ideas of what is 'natural' for bodies and for environments are often rooted in *reactions against official STM*; they may be as much the creations of popular Romanticism as of popularised science (Berridge, 1996; Shilts, 1987).

When atheists dispensed with God in the late nineteenth or early twentieth century, some thought that the human sciences would be sufficient guides for conduct and politics. But this was a rare position, for then as now, naturalistic accounts of man were not the only alternative to formal religion (Lepénies, 1985); the human condition and its meanings could be explored through culture, through the hermeneutic worlds of philosophy and literature, and by seeing in 'nature' a source of inspiration rather than information. To explore that strand of meaning, we conclude the chapter by returning again to 1800 and to the reactions against analytical science.

Nature and culture

Some nineteenth-century novelists regarded themselves as natural historians, as analysts of humanity or even as experimentalists, but other Victorian traditions seemed more at odds with 'science'. For example in Matthew Arnold's *Culture and Anarchy*, we see a familiar formalisation of 'two cultures': the opposition between the sciences that told you how to do worldly things, and the humanities through which you discovered what to do and why (Williams, 1958). This opposition derived from Germany, it was characteristic of many brands of idealistic philosophy, and it remains crucial to our intellectual world. But it is important to note that the opposition was and is chiefly between

'approaches' rather than the topics addressed. Both nature and human history could be read for facts and laws *or* for feeling and values; or, indeed, through the various combinations of these approaches which remained widespread in Western culture throughout the twentieth century.

In large measure, this opposition was a product of Romanticism, a complex set of reactions against the eighteenth-century rationalism which had pictured man as building up knowledge like a naturalist arranged plants, or as an analytical machine reducing complex phenomena to simple principles. The very success of 'analysis' in the decades around 1800 contributed to the Romantic reaction against 'dissection' and reductive explanations – though new types of 'analysis', notably morphology, could be thought of as anti-reductive (Cunningham and Jardine, 1990). Romanticism is indeed complex, but one key lies in the presumed relation of nature and man, described by Charles Taylor as follows: 'The meaning that natural phenomena bear is no longer defined by the order of nature itself or by the Ideas which they embody. It is defined through the effects of the phenomena on us, in the reactions they awaken. The affinity between nature and ourselves is now mediated not by an objective rational order but by the way that nature resonates in us' (Taylor, 1989: 299).

To take a widely known and influential example: in the philosophical poems of the English Romantic, William Wordsworth, the sense of nature as a teacher of morals is overwhelming. External nature resonates with the soul that is the nature in us at our birth, and for us to contemplate the hills and plants, or the lives of the people close to nature, is to cultivate one's moral self (Gill, 1989). In the later nineteenth century, John Ruskin, the art critic, painter and amateur geologist, developed these same sentiments and linked them to trenchant attacks on the ugliness of capitalism. At one level, theirs were powerful calls to the cultivation of a kind of 'natural history', and such sentiments have remained a major motivation of naturalists and conservationists – from the English Lake District which was home to both Wordsworth and Ruskin, to the national parks of America, and

now to global nature-tourism (Gould, 1988; MacKenzie, 1988). But Wordsworth (and Ruskin) drew the line at reductive analysis – ‘we murder to dissect’ – and that attitude, too, has echoed to our time.

We see it clearly in some of the feminist critiques of science from the mid-nineteenth century onwards, most literally in the anti-vivisection movement that first thrived in England around 1870, not least among women (French, 1975). We shall note in Chapter 5 that certain kinds of nineteenth-century science could be termed Romantic in their concern with unity and development rather than analysis into components. The ongoing debates around feminism and science pose questions about the plausibility and productivity of ‘alternative’ styles of science and about approaches to biology that would be ‘more respectful’ of organisms as wholes (Harding, 1991; Harding and O’Barr, 1987; Keller, 1983); and that same orientation is also evident in popular literature on the ‘holistic’ approaches to nature preferred by ‘ecologists’ and by critics of ‘reductionist’ medicine (Bramwell, 1989; Worster, 1994). Behind all these complex and political debates, we see the tensions between nature as the object of classification and analysis, and nature as ‘whole’ and meaningful; we see the boundaries moving and blurred, always in dispute. Sometimes natural-historical knowings were contrasted with analytical methods, and sometimes ‘natural history’ was itself contested – between those who stressed classification and order and those who stressed meanings and individuals.

And this was also the case for human history – was it an extension of natural history (or even a field for analysis), was it to be searched for evidence of God’s providence or for laws, or was it to be contemplated as a means of understanding other ages ‘from the inside’? To what extent could these goals be compatible? These questions were crucial for nineteenth-century configurations of science and scholarship; they remain important for us (Bowler, 1989b; Levine, 1986; Mandelbaum, 1971; Outhwaite, 1975).

History here, of course, includes the history of STM, and the texts that can be ‘read’ hermeneutically include the papers and

books of past or present 'scientists'. *If* these are read primarily for the construction of meanings, then they become part of a hermeneutic history. In such ways the 'nature' presented by STM becomes a reflection, not just of God or of how *things* really are, but of the investigators who produced the texts, and of the cultures which shaped them. R. G. Collingwood, who had read 'old' science books as a child in John Ruskin's library, realised that history could be regarded as prior to the 'construction' of nature. Recent history of STM follows this insight, to discover how all knowledges have been created within systems of meaning over time. So here we return to the themes of the start of this chapter; here, indeed, is the place of this book in my account of Western culture. I use historical analysis to reconstruct the many 'projects' and 'understandings' in the past and present of STM; history becomes part of our reflections on nature and science.

With that circle completed, we turn now to focus on the histories of the ways of knowing that comprise modern STM. We turn from the writing of human history to the making of natural history – and then to analysis, experiment and technoscience – but always trying to see the projects that constituted modern STM as also constitutive of our human histories.

Notes

- 1 But I do not mean to suggest the opposite mistake – an eternal 'natural-historical' language for patients and symptoms. Clearly, some aspects of technical medicine are domesticated in patients' lives as drugs and/or as concepts. Just how they are understood and used is not well recorded – we need more studies of the hermeneutics and natural history of life under technoscience. See Cornwell (1984) for one model, or chapter 7 of Collins and Pinch (1998). Perhaps some of the tools developed by historical sociologists for analysis of interactions between professional groups could be applied here. See, for example, Star and Griesmer (1989) on boundary objects, or Peter Galison (1997) on 'creole' languages in physics. See also Pickstone (1994b) for the various material cultures of medicine, and Miller (1991) for modern material cultures more generally.

- 2 To see how good that teacher is, read Bray (1997).
- 3 See also Gouk (1997).
- 4 See the two excellent surveys of science and religion by Barbour (1966) and Brooke (1991).